Group Project Milestone 1:
Literature Review and Exploratory Data Analysis

Predicting Delivery Performance and Customer Satisfaction

An analysis of 99,441 orders from the Olist Brazilian e-commerce marketplace

Group 10

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Dashboard: `https://it5006-group-project-10.streamlit.app`

Repository: `https://github.com/gengyuzhu/IT5006-Group-Project-10`

1 Literature Review

1.1 Scope

This project analyses the *Olist Brazilian E-Commerce* dataset: nine relational tables covering 99,441 orders placed on a Brazilian marketplace between September 2016 and October 2018 [15]. Olist aggregates small sellers onto Brazil's large retail platforms and handles fulfilment, so each order links a customer, one or more sellers, a basket, a payment, a delivery and usually a review – a structure supporting all three question families the brief identifies: operational performance, customer experience and commercial outcomes. The review below covers what Phase 2 depends on: fulfilment and satisfaction, model families for tabular data, feature engineering on a relational order model, and metrics that survive severe imbalance.

1.2 Fulfilment, delivery promises and customer satisfaction

The e-commerce service-quality literature is unusually direct about which lever matters. Parasuraman et al. [16] decompose electronic service quality into efficiency, *fulfilment*, system availability and privacy, and find fulfilment – the accuracy of delivery promises and the condition in which goods arrive – among the strongest contributors to perceived quality. Rao et al. [18] sharpen this into logistics terms across 260 online retailers. Transactional evidence at platform scale now confirms it: across more than 15 million marketplace orders, Deshpande and Pendem [6] trace a chain rather than a single link – logistics performance drives the service rating a customer leaves, and that rating in turn moves seller sales. That chain makes a low-review model a commercial instrument rather than a diagnostic.

The promise itself is a second, less obvious decision variable. Cui et al. [5] show that committing to a faster delivery date raises sales but also increases returns and depresses retention, so the optimal promise trades demand against the cost of breaking it. An estimated delivery date is therefore a business decision, not a forecast, and should not be expected to behave like one (Section 2.6). Prediction likewise pays only when it feeds a decision: Kandula et al. [11] route machine-learned delivery outcomes into a scheduling optimiser and report cost savings of up to 10.2%.

Delivery is thus the dominant *observable* input to how a customer rates a transaction, which is why our second candidate problem is framed after delivery rather than at checkout. The same literature warns what is *not* observable here – product expectations, seller communication, returns handling – none of which appears in an order table, and any performance ceiling we report must be read against that omission.

1.3 Model families for tabular transactional data

For structured data of this size the practical model space is well characterised. Regularised linear and logistic models are the natural baseline: cheap, interpretable, and – as Section 2.6 shows – revealing about distribution shift. Tree ensembles are the usual step up, whether by bagging decorrelated trees [2] or fitting residuals stagewise with shrinkage [4, 7]. Grinsztajn et al. [9] benchmark both against modern deep architectures across 45 datasets and find tree-based methods remain state of the art at this scale, the gap owing to inductive biases better matched to irregular tabular targets.

We therefore adopt a small model budget: one linear family (ridge and ℓ_2-regularised logistic regression) and one tree ensemble (histogram-based gradient boosting), reused across both framings. Section 2.6 justifies that complexity in both directions – the ensemble earns its place on lead time, while on repeat purchase neither family beats the baseline.

1.4 Feature engineering on a relational order model

Deriving features from nine tables raises a problem a flat file does not. Three of them – items, payments and reviews – are one-to-many against orders, so a naive join multiplies rows and reweights every aggregate towards multi-item orders. The remedy is to aggregate each child relation to the analysis grain before joining. Beyond that the usual transactional vocabulary applies: basket counts and sums, ratios that normalise for order size, calendar encodings, and geographic distance from postcode centroids. High-cardinality categoricals such as the 74 product categories suit target encoding [14] better than one-hot, provided it is fitted inside the cross-validation loop.

Leakage is the dominant risk. Kaufman et al. [13] formalise it as information about the target unavailable at prediction time, and identify the two forms that threaten this dataset: post-outcome features and improper partitioning. Kapoor and Narayanan [12] show how routinely both survive peer review, cataloguing eight types of leakage across 294 affected papers in 17 fields; ours are their *illegitimate feature* and *no clean separation between training and test data* categories, and we guard each with an explicit assertion rather than a convention. Critically, this framing makes leakage relative to a *prediction moment* – the principle our feature policy is built on, since delivery outcomes are forbidden when forecasting delivery but admissible when predicting a review not yet written. On partitioning, Bergmeir and Benítez [1] show that standard k-fold cross-validation is invalid for temporally ordered data.

1.5 Evaluation under imbalance and drift

Both classification targets here are heavily imbalanced, and the brief is explicit that accuracy alone misleads. Saito and Rehmsmeier [19] show that precision–recall curves are more informative than ROC curves under strong imbalance, because ROC-AUC is insensitive to a large true-negative mass. We report PR-AUC against its no-skill floor alongside balanced accuracy, treating raw accuracy as a diagnostic rather than a headline.

The obvious next step – correcting the imbalance – is one we deliberately do not take. van den Goorbergh et al. [20] test random under-sampling, over-sampling and SMOTE [3] and find none improves discrimination, while all three badly miscalibrate the resulting probabilities, systematically overestimating minority-class risk; they recommend adjusting the decision threshold on uncorrected data instead. We therefore do not resample. We do weight the loss by class, which serves the same end as their threshold shift, but it carries the same caveat: we treat model outputs as *decision scores rather than calibrated risks*. Should Phase 2 need calibrated risk – to rank at-risk orders by expected cost – we will fit on the observed prior and tune the threshold, reporting calibration explicitly.

Regression brings a second complication. Hyndman and Koehler [10] argue for judging errors against a naive benchmark, decisive here because delivery performance improves dramatically across the window (Section 2.3). Under such a level shift, R^2 against a baseline fitted on the past turns negative even for a useful model, whereas mean absolute error stays interpretable. This is textbook concept drift [8], whose recommendation that non-stationary models be refitted with respect to time shapes our Phase 2 plan. All modelling uses `scikit-learn` pipelines [17], keeping preprocessing inside each fold.

1.6 Conclusion: trends, gaps and opportunities

Three themes emerge. The service-quality literature converges on *fulfilment* as the dominant observable driver of e-commerce satisfaction, and now traces it through to seller revenue. The tabular machine-learning literature converges on a small set of model families, with tree ensembles the reasonable ceiling at this scale. And the methodological literature is emphatic – increasingly

so – that on imbalanced, temporally ordered data the choice of split, metric and imbalance handling decides whether a number means anything at all.

The gap we address is narrow but concrete. Most work on this marketplace is descriptive, or predicts review scores from features that would not exist when the prediction is needed. Little of it is explicit about *when* a model would run – a choice that changes both what is admissible and what is achievable, by a factor of two in our own results. Making the prediction moment a first-class design decision, and measuring its cost, is the contribution of Section 2.6.

2 Exploratory Data Analysis

2.1 Dataset overview and data model

The dataset comprises **nine** related CSV tables totalling roughly 1.45 million rows. `orders` is the spine, at 99,441 rows – one per order – and the other eight attach to it directly or through `order_items`. Figure 1 shows the recovered structure and Table 3 gives the full data dictionary.

Three tables are **one-to-many** against orders, and joining them naively is the first trap the data sets: a direct orders–items–payments merge yields 118,437 rows rather than 99,441, a 19% inflation that silently reweights every mean towards large orders. We aggregate each child table to one row per order first, then join, asserting the grain afterwards.

The second trap is identity. `customer_id` is regenerated for every order, so it is an order key, not a person: 99,441 distinct `customer_id` values map to only **96,096** distinct `customer_unique_id` values. Grouping on the wrong one reports zero repeat buyers; the right one finds 2,997 people (3.1%) who ordered again, one of them 17 times. We carry `customer_unique_id` throughout and split on it (Section 2.6).

2.2 Data quality

Table 1 records every issue and our decision; the decisions live in `src/data_load.py`, so the code and this table cannot drift apart.

Missingness here is **structural rather than random**. 97.0% of orders reach status *delivered*; the rest are shipped, cancelled, unavailable or still processing, and 2.98% therefore carry no delivery timestamp. That is not a defect to impute – the event has not happened – so delivery targets are undefined there and excluded from delivery models. Elsewhere the data is remarkably complete: 99.2% of orders carry a review and product attributes are missing for under 2% of products.

Two decisions shape the window. The first four months are a pilot contributing 326 orders against a 2018 run rate above 6,000 per month, and the final weeks are truncated mid-cycle, so we restrict the analysis to **2017-01-01 to 2018-08-31**, leaving 99,091 orders (Figure 2).

2.3 Temporal patterns: growth, seasonality and drift

Monthly order volume roughly triples across the window, peaking at **Black Friday, November 2017** with 7,288 orders, about 1.5 times a typical month (Figure 3a). Purchasing is weekday-heavy, peaks at 16:00, and is 22% weekend (Figure 4).

The most consequential temporal finding is not seasonality but **drift**. Mean lead time falls from 16.9 days in February 2018 to 7.7 in August 2018 – Olist materially improved fulfilment over twenty months (Figure 3b) – and across the chronological split used later the mean falls by **5.34 days** between training and test.

Lateness behaves differently: it is **volatile rather than trending**, swinging from 1.4% (June 2018) to 21.4% (March 2018). Those spikes align with Black Friday congestion and the

nationwide truckers' strike that paralysed Brazilian road freight in May 2018 – exogenous shocks no order-level feature available at checkout can anticipate.

Together these facts set the evaluation strategy. Under a level shift this size, R^2 against a baseline fitted on the past is negative even for a genuinely useful model, so we report **mean absolute error** as the headline regression metric [10], and treat periodic refitting as a Phase 2 requirement rather than an optimisation [8].

2.4 Customers, sellers, products and geography

The marketplace is geographically **lopsided**, and that asymmetry drives the delivery problem: **59.7%** of sellers sit in São Paulo state against **42.0%** of customers, only 35.7% of orders stay within the customer's own state, and the median customer–seller distance is **434 km** (Figure 5). Distance is the strongest checkout-time correlate of lead time ($r = 0.40$), ahead of the platform's own estimate ($r = 0.38$).

Demand is long-tailed across 74 categories, the top ten covering two-thirds of orders. The typical order is small – median item value R$86.90 – but **freight is not a rounding error**, a median R$17.17 or 22% of item value. Payment is characteristically Brazilian: 75.4% credit card, 51.4% in instalments (Figure 6).

2.5 Reviews, and the relationship that dominates

Review scores are severely imbalanced – **77.1%** are four or five stars – exactly the imbalance the brief warns about. The interesting structure is what produces the minority.

A late delivery raises the probability of a 1–2 star review from **9.2%** to **54.0%**, a **5.9-fold** increase, monotone in how late the parcel is (Figure 7). This is the clearest relationship in the dataset, and it reproduces at order level the logistics–rating link Deshpande and Pendem [6] establish across 15 million orders – where it carries through to seller sales. It also has a methodological consequence: a model predicting review score *at checkout* cannot see the variable that matters most.

2.6 Candidate problem exploration

We examined three candidates. All probes use `scikit-learn` pipelines on a chronological split (train to 2018-04-30, validation to 2018-06-30, test thereafter) additionally **grouped on** `customer_unique_id`, removing the 227 buyers who would otherwise appear on both sides.[a]

Leakage depends on when the model runs. Rather than one global rule we define a forbidden set per **prediction moment**. *At checkout* nothing about fulfilment or the customer's opinion exists: delivery dates, review scores and even `order_approved_at` are future. *At delivery* the parcel has arrived and we are predicting a review not yet written, so the delivery outcome is history and admissible – the brief itself lists "delivery performance" among the features for this problem. The guard `assert_no_leakage(features, regime)` enforces the right set, and every modelling cell must pass it.

The cost of getting this wrong is measurable: adding one forbidden column, the realised delivery time, lifts the late-delivery model's ROC-AUC from **0.589 to 1.000** (Figure 9). That is not a result; it is the target restated.

[a] **Note on splitting.** We split by time rather than by stratified random sampling because the target drifts (Section 2.3) and a random split lets a model learn from the future. Chronology alone is insufficient though: 2,997 people placed more than one order, so a temporal cut can still put one of a person's orders in training and another in test. This is the concrete form of the brief's `customer_id` versus `customer_unique_id` warning, and why we also report a test subset of customers never seen in training.

Candidate 1 — delivery performance, framed twice (primary). One question answered as a regression on lead time and a classification of late versus on-time, satisfying the brief's requirement for both task types from one coherent problem. Stakeholder: operations.

The regression works. Gradient boosting reaches **MAE 3.89 days** against 6.65 for a constant fitted on the training period, a **41.5%** reduction, with ridge at 4.61 days in between. Plain least squares, the simpler starting point, is unusable here – the design matrix is rank-deficient, so its coefficients explode – and ridge at $\alpha = 1$ is the minimum regularisation that repairs it. Notably, Olist's own delivery promise is a *worse* point forecast than the constant (MAE 9.79 days) – expected rather than anomalous, since a promised date is a commercial decision, padded because a broken promise costs returns and retention [5]. A useful feature, not a useful prediction.

The classification is weak, and Section 2.3 explains why: balanced accuracy 0.580, PR-AUC 0.121 against a 0.074 floor. Crucially, **always predicting "on time" scores 0.926 accuracy** while catching not a single late order – the exact trap the brief describes.

Candidate 2 — low review score, predicted at delivery (secondary). The strongest classifier we found: PR-AUC **0.402** against a 0.110 floor (3.6×), balanced accuracy **0.692** – and identical on customers never seen in training (PR-AUC 0.403), so it is not memorising buyers. The prediction moment is what makes it work: the same target predicted *at checkout* reaches only PR-AUC 0.212. Stakeholder: customer experience, with a concrete action – flag an at-risk order for outreach before the review lands.

Candidate 3 — repeat purchase within 30 days (rejected). This fails twice over. The base rate is only 1.8% – 96.9% of customers appear exactly once – and the probe reaches balanced accuracy 0.517, essentially chance. Separately, **right-censoring destroys the evaluation**: at the more natural 90-day horizon *no* labelled orders remain in the test window at all, a structural limit of a 21-month extract that no better model repairs. Table 2 scores all three.

2.7 Key insights

1. Three of the nine tables are one-to-many against orders; a naive join inflates the data 19% and biases every average towards large orders.
2. `customer_id` is an order key, not a person – the difference between reporting 0% and 3.1% repeat buyers.
3. Delivery performance **drifts**: lead time halves over the window, so MAE rather than R^2 is the honest regression metric.
4. Lateness is **volatile, not predictable** (1.4–21.4% monthly), which bounds the late-delivery classifier near 1.6× the no-skill floor.
5. A late delivery raises the 1–2 star rate from 9.2% to 54.0%, the strongest relationship in the data.
6. **The prediction moment is a design decision, not a detail**: it nearly doubles achievable PR-AUC one way, and inflates ROC-AUC to a meaningless 1.000 the other.

We expect a tuned Candidate 2 classifier to reach balanced accuracy in the high 60s, and Candidate 1's regression MAE of 3.5–4 days. The largest driver of both, month-to-month carrier conditions, is not observable at prediction time; a late-delivery classifier reporting 95% accuracy would be the majority-class rule or the leaked variant above. The dashboard and repository are linked on the cover page; both read through the same `src/data_load.py` entry point as this report, and `src/report_stats.py` regenerates every number quoted here.

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A Data Model, Audit and Dictionary

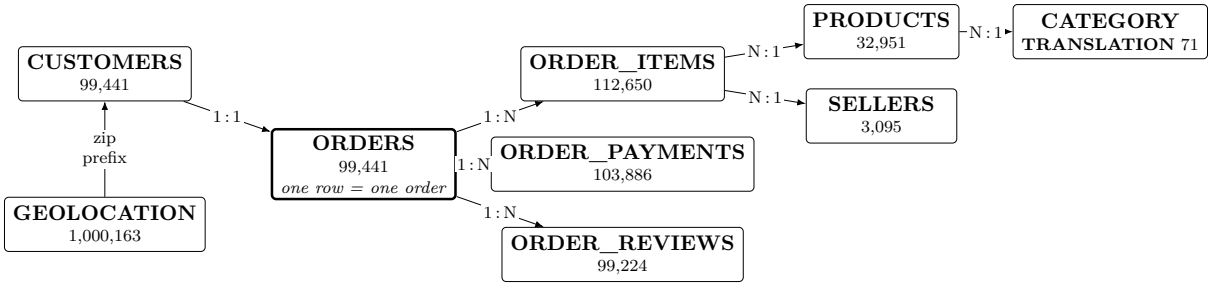


Figure 1: The nine source tables. `ORDERS` is the analysis grain. The three one-to-many children (bold arrows to items, payments and reviews) are each aggregated to one row per order *before* joining; merging them directly inflates the table by 19% and biases every average.

Table 1: Data quality audit and cleaning decisions, implemented as `CLEANING_DECISIONS` in the data-loading module.

Issue	Evidence	Decision
One-to-many joins	112,650 items and 103,886 payments against 99,441 orders; a naive merge yields 118,437 rows	Aggregate each child table to one row per order, then join; assert the grain
Customer key	99,441 `customer_id` values but 96,096 `customer_unique_id`; 2,997 people ordered more than once	Carry `customer_unique_id` and group on it for repeat behaviour and for splitting
Undelivered orders	2.98% of orders have no delivery timestamp; status is *delivered* for 97.0%	Delivery targets computed only for delivered orders; the rest are `NaN`, never imputed
Review keys	`review_id` is not unique (789 ids reused); 547 orders carry 2–3 reviews and 768 carry none	Key on `order_id`, keep the earliest review; no review means `NaN`
Sparse review text	`review_comment_title` present for 11.7% of reviews, `review_comment_message` for 41.3%	Use `review_score` only; coverage does not support NLP features
Geolocation grain	1,000,163 rows for 19,010 zip prefixes, including points outside Brazil	Collapse to a median coordinate per prefix; use it for great-circle distance
Missing product attributes	610 products (1.9%) have no category	Label `unknown` rather than dropping orders; impute numerics in-pipeline
Coverage ramp	326 orders before 2017; final weeks truncated mid-cycle	Restrict to 2017-01-01 .. 2018-08-31 (99,091 orders)
Portuguese categories	71 category names	Translate to English for all output; keep the original key

Table 2: Candidate problems against the brief's scoping checklist.

Criterion	**C1 delivery (checkout)**	**C2 low review (delivery)**	**C3 repeat (checkout)**
Derivable target	Yes, from order timestamps	Yes, `review_score` ≤ 2	Yes, next order by person
No leakage	Yes, checkout regime	Yes, at-delivery regime	Yes, checkout regime
Sufficient signal	Regression yes (-41% MAE); classification weak ($1.6\times$)	Yes, PR-AUC 0.40 vs 0.11 ($3.6\times$)	No, $2.0\times$ at a 1.7% base rate
Manageable imbalance	8.1% late; loss weights and PR-AUC, no resampling	14.6% low; same handling	2.6%, too extreme
Clear stakeholder	Operations	Customer experience	CRM
Verdict	**Primary**	**Secondary**	Rejected

B Figures

Table 3: Data dictionary for the nine source tables, in the column-level format used by the NIBRS reference dictionary supplied with the brief. *Nullable* is measured, not declared: Y means the column actually contains missing values in this extract, with the observed rate given.

Table	Column	Type	Null	Comment
orders	order_id	text	N	Primary key. The analysis grain
	customer_id	text	N	**Per-order** key, not a person
	order_status	text	N	8 values; *delivered* for 97.0%
	order_purchase_timestamp	datetime	N	The checkout prediction moment
	order_approved_at	datetime	Y 0.16%	Post-checkout; excluded as a feature
	order_delivered_carrier_date	datetime	Y 1.79%	Post-outcome
	order_delivered_customer_date	datetime	Y 2.98%	Post-outcome; defines both delivery targets
	order_estimated_delivery_date	date	N	Promised at checkout; **admissible**. Date only, 459 distinct values
order_items	order_id	text	N	**1:N** — 112,650 rows for 98,666 orders
	order_item_id	int	N	Sequence within the order, 1–21
	product_id, seller_id	text	N	Foreign keys
	shipping_limit_date	datetime	N	Post-checkout; excluded
	price, freight_value	float	N	Per item; summed to order level
order_payments	order_id	text	N	**1:N**. One order has no payment row
	payment_sequential	int	N	1–29
	payment_type	text	N	5 values; credit card 75.4%
	payment_installments	int	N	0–24; > 1 for 51.4% of orders
	payment_value	float	N	Summed to order level
order_reviews	review_id	text	N	**Not unique** — 789 ids reused across orders
	order_id	text	N	**1:N**; 547 orders carry 2–3 reviews
	review_score	int	N	1–5; target of Candidate 2
	review_comment_title	text	Y 88.3%	Too sparse to use
	review_comment_message	text	Y 58.7%	Too sparse to use; no NLP features
	review_creation_date	date	N	Post-outcome for the review target
	review_answer_timestamp	datetime	N	Post-outcome
customers	customer_id	text	N	Joins 1:1 to orders
	customer_unique_id	text	N	**The actual person**; 96,096 of them. Grouping key for splits
	customer_zip_code_prefix	int	N	Joins to geolocation
	customer_city, customer_state	text	N	4,119 cities, 27 states
sellers	seller_id	text	N	3,095 sellers
	seller_zip_code_prefix	int	N	Joins to geolocation
	seller_city, seller_state	text	N	59.7% of sellers are in SP
products	product_id	text	N	32,951 products
	product_category_name	text	Y 1.85%	Portuguese; 73 values. Nulls labelled *unknown*
	product_weight_g	float	Y 0.01%	Summed to order level
	product_length/height/width_cm	float	Y 0.01%	Combined into item volume
	product_photos_qty	float	Y 1.85%	Listing-quality proxy
geolocation	geolocation_zip_code_prefix	int	N	19,015 prefixes over 1,000,163 rows
	geolocation_lat, _lng	float	N	Collapsed to a median per prefix; out-of-Brazil points dropped
category_ translation	product_category_name	text	N	Join key
	product_category_name_english	text	N	71 translations
Derived in src/ — not present in any source file:				
derived	delivery_days	float	Y	Purchase to arrival; target of C1A
derived	is_late	bool	Y	Arrival after the promised date; target of C1B
derived	distance_km	float	Y 1.2%	Great-circle customer–seller distance
derived	estimated_days	float	N	Promised window in days
derived	lead_category	text	N	Category of the most expensive item
derived	freight_ratio	float	Y	Freight as a share of item value

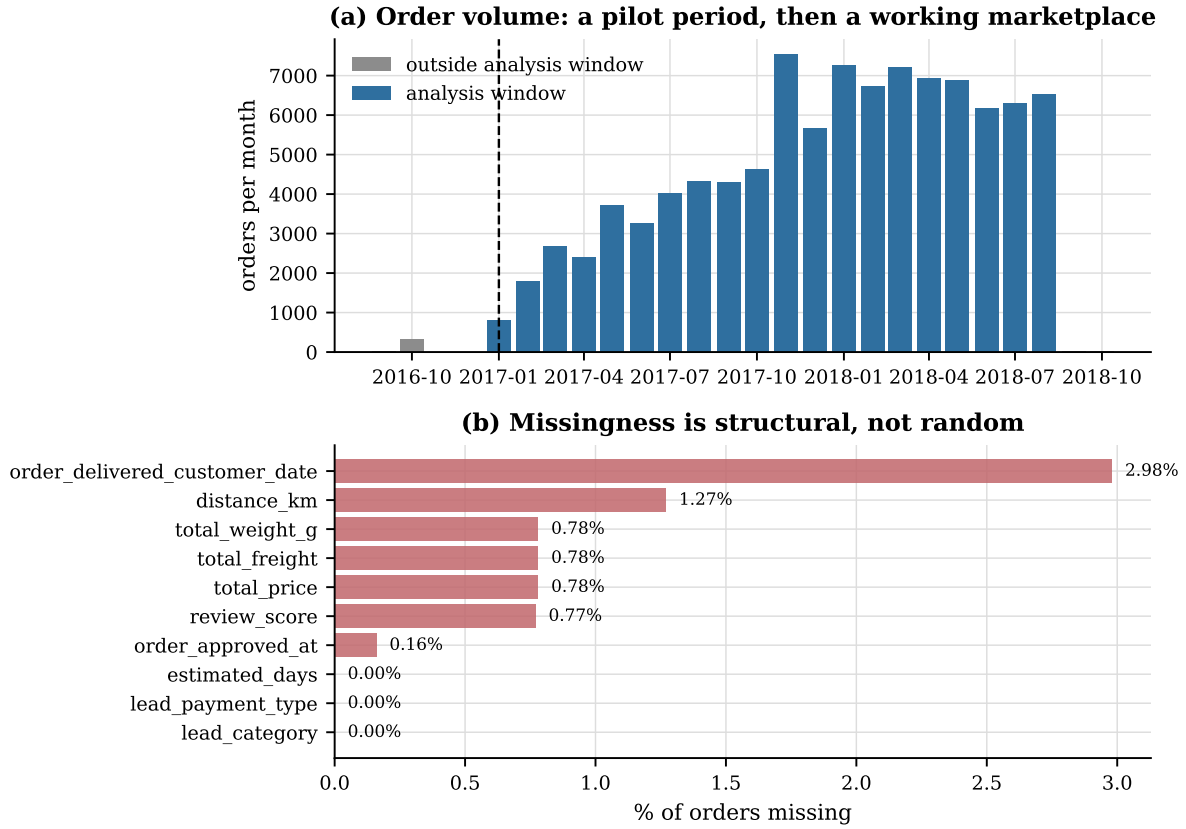


Figure 2: (a) Monthly order volume. The pilot period before 2017 contributes 326 orders in total and is excluded from the analysis window. (b) Missingness by column: structural rather than random, dominated by orders that never reached the customer.

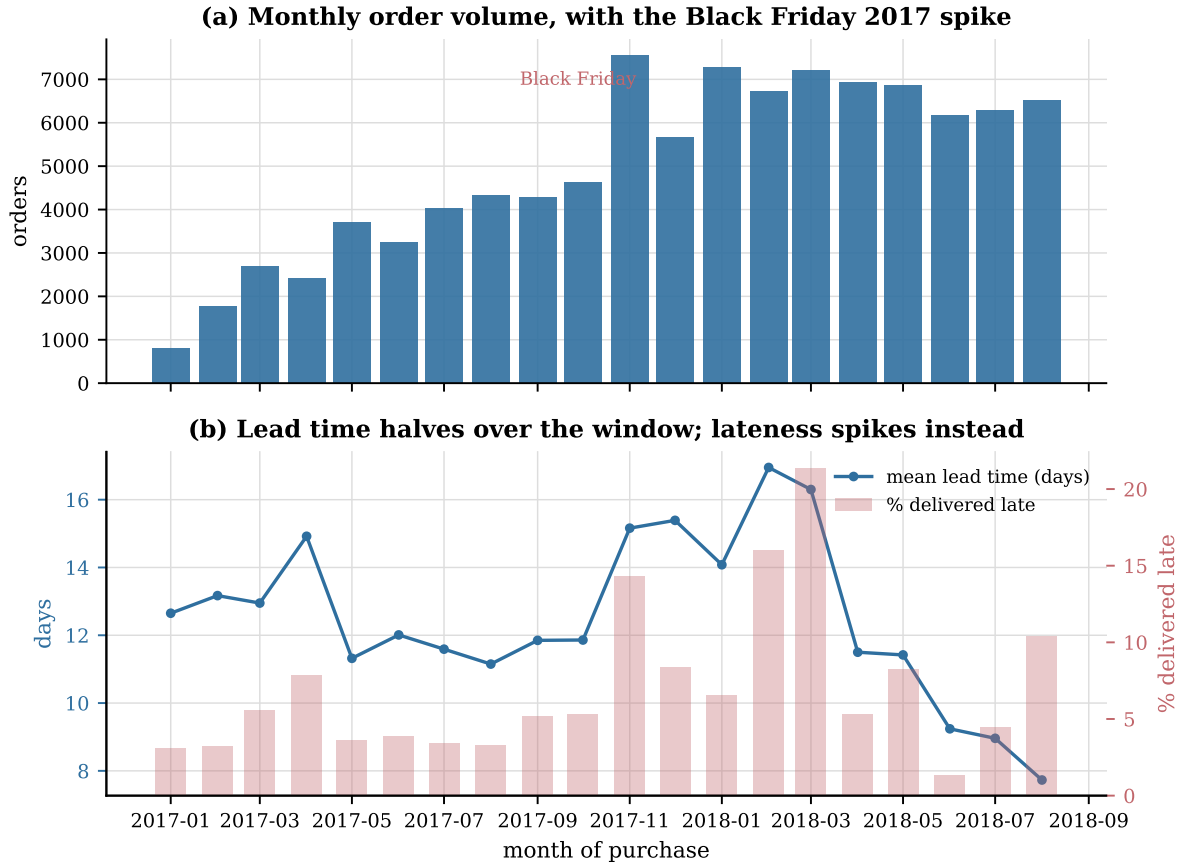


Figure 3: (a) Order volume with the Black Friday 2017 spike. (b) The central temporal finding: mean delivery lead time halves across the window (line, left axis) while the late rate is volatile rather than trending (bars, right axis), spiking at Black Friday and during the May 2018 truckers' strike.

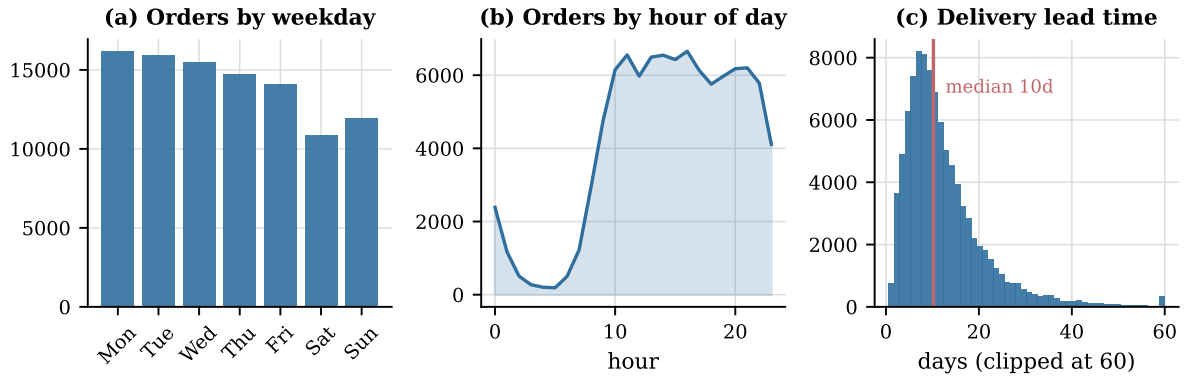


Figure 4: Purchase rhythm and the lead-time distribution. Ordering is weekday-heavy and peaks at 16:00; lead time is right-skewed with a median of 10.2 days and a long tail.

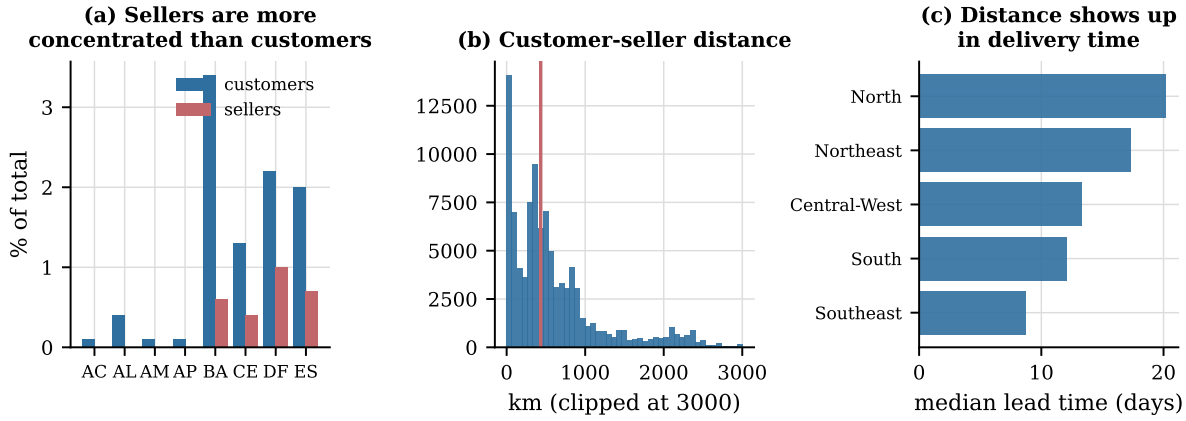


Figure 5: (a) Sellers are far more concentrated in São Paulo than customers are, so most orders must cross the country. (b) Customer–seller distance. (c) That distance shows up directly in regional delivery times.

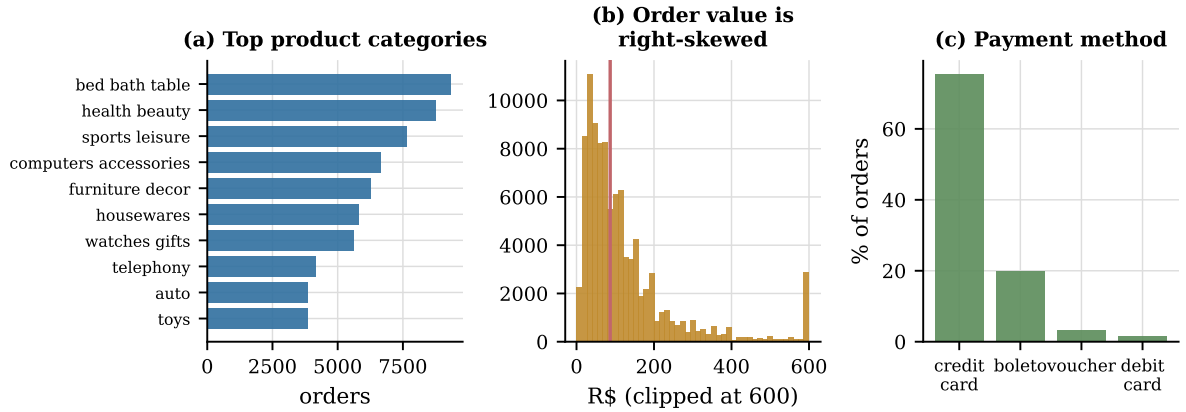


Figure 6: (a) Demand is long-tailed across 74 categories. (b) Order value is strongly right-skewed. (c) Credit card dominates, usually in instalments.

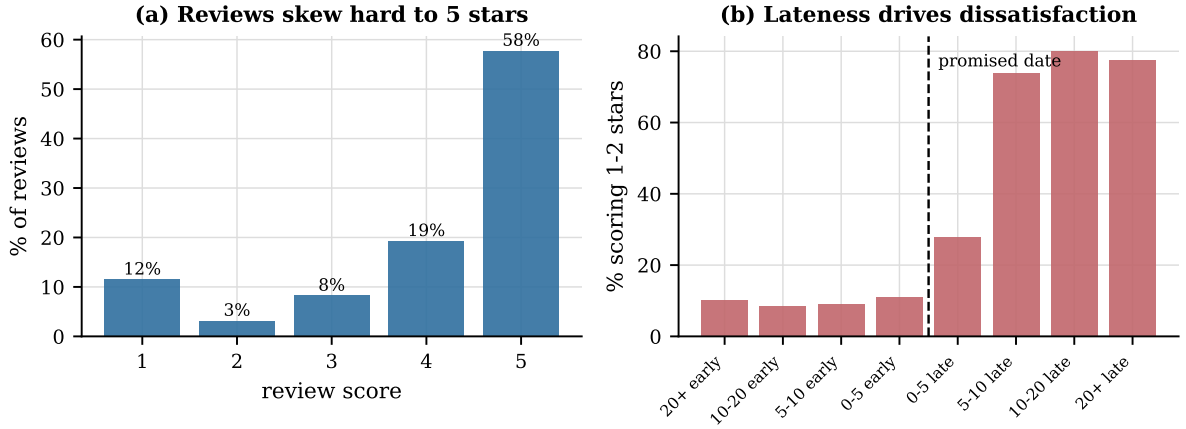


Figure 7: (a) Reviews are severely imbalanced towards five stars. (b) The strongest relationship in the dataset: the 1–2 star rate rises monotonically with lateness, from 9.2% for on-time orders to 54.0% for late ones.

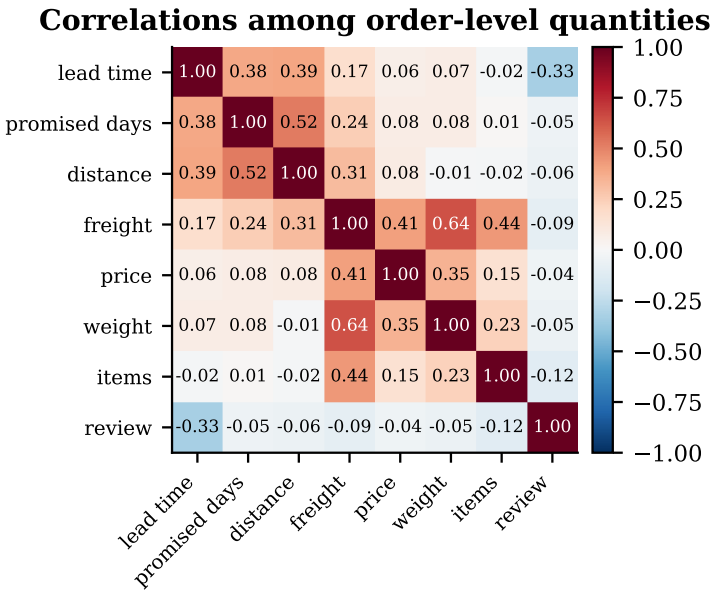


Figure 8: Correlations among order-level quantities. Distance and the platform's own promise are the strongest checkout-time correlates of lead time.

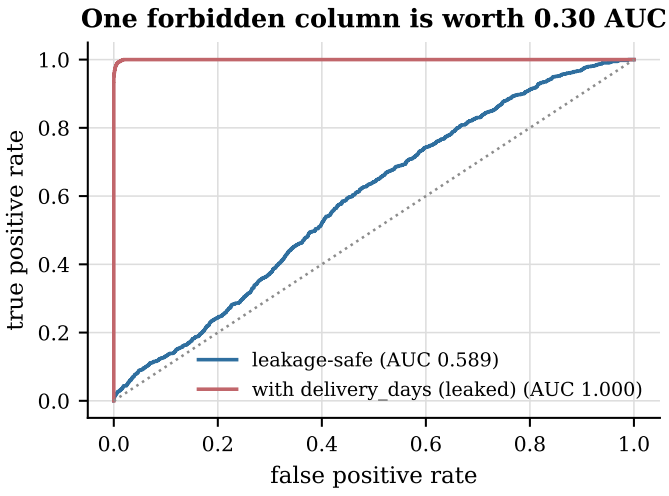


Figure 9: What one forbidden column is worth. Adding the realised delivery time to the late-delivery model takes ROC-AUC from 0.589 to 1.000 — the model is reading the answer, not predicting it.

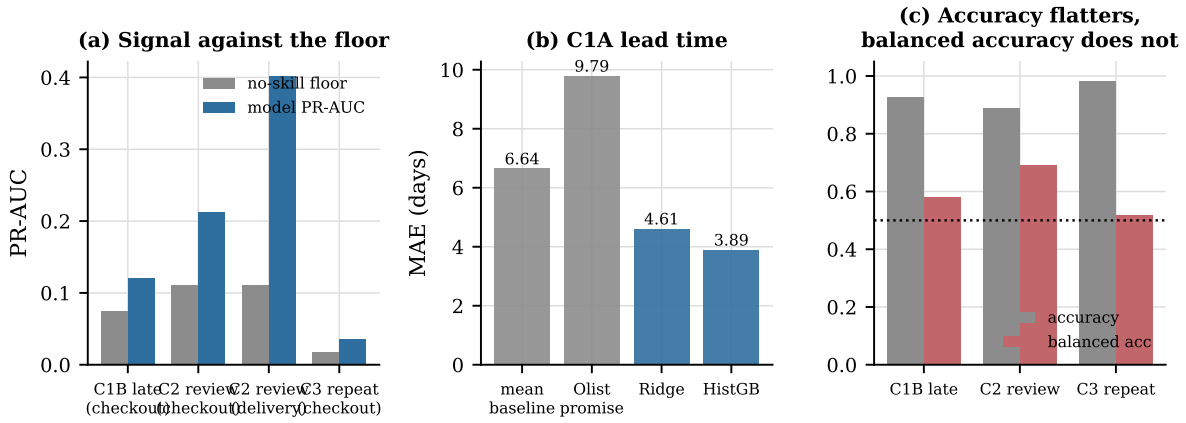


Figure 10: Candidate comparison. (a) PR-AUC against the no-skill floor for each candidate and prediction moment. (b) Lead-time MAE: the platform's own promise is worse than a constant. (c) Raw accuracy flatters every imbalanced target; balanced accuracy does not.

AI Tool Declaration

We used OpenAI's ChatGPT and Anthropic's Claude to generate and validate ideas, improve expression and refine our writing. We also used the models to debug our code. All analytical decisions, results and conclusions were reviewed and verified by the team against the underlying data, and we are responsible for the content and quality of the submitted work.